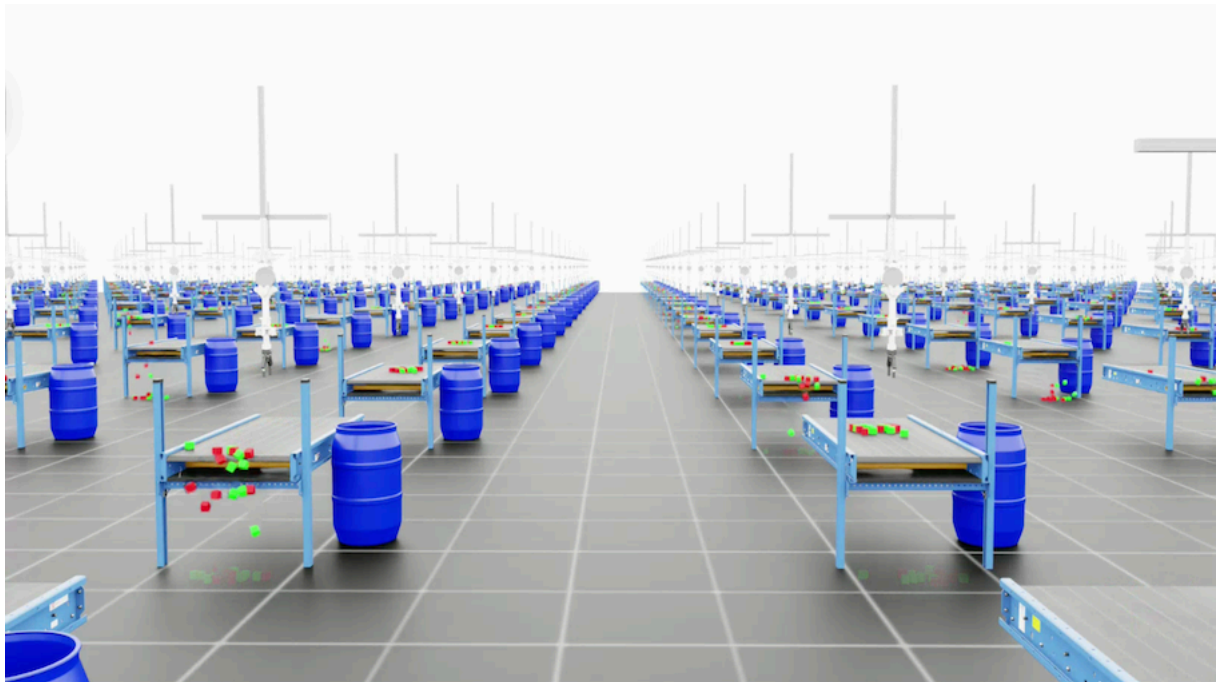


Multi-Agent Reinforcement Learning for Condition-Based Textile Sorting with a Tendon-Driven and Collaborative Robot System

Master Thesis in the program Computational Engineering M. Sc.

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List of Abbreviations

DoF	Degree of Freedom
FEM	Finite Element Method
GPU	Graphics Processing Unit
MDP	Markov Decision Process
MPM	Material Point Method
PBD	Position-Based Dynamics
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
ROS	Robot Operating System
VBD	Vertex Block Descent
XPBD	Extended Position-Based Dynamics

1 Introduction

1.1 Motivation and Context

The global fashion and textile industry generates substantial volumes of post-consumer waste, with a large fraction ending up in landfill or incineration rather than being recovered for reuse or recycling. Within the European Union, regulatory pressure to increase textile recycling rates is growing, with separate collection of textile waste becoming mandatory in member states. A key bottleneck in the recycling pipeline is the sorting step: incoming textile waste must be classified by material composition, garment type, and condition (e.g., intact, minor damage, major damage) before it can be routed to appropriate downstream processes such as fiber-to-fiber recycling, downcycling, or second-hand resale. Currently, this sorting is performed predominantly by manual labor, which is slow, ergonomically demanding, and difficult to scale. Automating textile sorting with robotic systems is a promising approach to increasing throughput and consistency, but it presents significant technical challenges. Garments are deformable objects that exhibit highly variable shapes, complex self-interactions, and contact dynamics that differ fundamentally from the rigid-object manipulation scenarios for which most industrial robots are designed [8]. Reliable textile sorting requires not only picking individual items from a mixed waste stream but also manipulating them into configurations that allow visual inspection—for example, stretching a garment flat to expose damage—and then routing each item to the correct output bin based on the inspection result. This sequence of pick, inspect, and sort naturally involves multiple manipulation primitives and can benefit from coordination between multiple robot arms operating on the same workpiece.

RL offers a data-driven approach to learning such complex manipulation behaviors. Rather than manually designing controllers for each manipulation primitive, RL enables policies to be discovered through trial-and-error interaction with the task environment, guided by a reward signal that encodes the desired outcome [7, 26]. When extended to *multi-agent* settings, RL can learn coordinated policies for teams of robots that must share a workspace, hand off workpieces, and jointly optimize a system-level objective.

Modern Graphics Processing Unit (GPU)-accelerated physics simulators have become essential enablers for this approach. By running thousands of environment replicas in parallel, they provide the sample throughput required for stable policy learning while avoiding the cost and risk of training on physical hardware [11]. NVIDIA Isaac Sim, with its PhysX 5 physics backend and the associated IsaacLab framework, provides an integrated toolchain that combines high-fidelity rigid-body and deformable-object simulation with a modular Markov Decision Process (MDP)-based task interface suitable for large-scale RL training [19, 22]. Recent developments in the Newton physics engine further extend this capability to realistic cloth simulation using Vertex Block Descent (Vertex Block Descent (VBD)) [3, 17], making simulation-based development of cloth manipulation policies increasingly practical.

This thesis is the second part of a two-part project conducted at the Institute for Factory Automation and Production Systems (FAPS) at Friedrich-Alexander-Universität Erlangen-Nürnberg. The preceding project thesis (Meyer, 2026) established a validated simulation of the 5-Degree of Freedom (DoF) tendon-driven robot and demonstrated RL-based control for reach and rigid-object place tasks in Isaac Sim. The present work extends that foundation to the full textile sorting pipeline: integrating deformable cloth simulation, adding a second collaborative robot arm, implementing camera-based damage classification, training coordinated multi-agent policies, and preparing the system for sim-to-real transfer.

1.2 Problem Statement and Objectives

Condition-based textile sorting from mixed waste streams poses a multi-faceted robotic manipulation problem that goes substantially beyond standard pick-and-place. The task requires a system that can (i) selectively retrieve target garments (e.g., T-shirts) from a moving conveyor while ignoring non-target objects, (ii) present each garment in a configuration suitable for visual inspection (e.g., stretched flat), (iii) classify the garment’s condition through camera-based analysis, and (iv) route the item to the appropriate output bin. Each of these sub-tasks involves contact-rich interaction with deformable objects whose behavior is difficult to model analytically [8], and the full pipeline requires coordination between multiple robotic agents operating in a shared workspace.

The robotic system considered in this thesis consists of two manipulators. The first is the tendon-driven 5-DoF pick robot validated in the project thesis: a 2-DoF prismatic positioning stage, a 3-DoF tendon-driven arm, and a Robotiq 2F-140 gripper, based on the design by Klein [6]. The second is a collaborative sorting arm that receives the garment from the pick robot, stretches it for camera-based inspection, and deposits it into a condition-specific bin. The two robots must coordinate their actions—timing the handoff, avoiding collisions, and jointly optimizing the sorting throughput and accuracy—which naturally frames the problem as a multi-agent RL task. A critical technical challenge is the simulation of cloth dynamics with sufficient fidelity for policy learning. Deformable cloth introduces a high-dimensional state space (many particles/vertices), complex self-collisions and folding behavior, and sensitivity to material parameters such as bending stiffness, friction, and damping [8, 12]. Policies trained on inaccurate cloth models risk exploiting simulation artifacts that do not transfer to real textiles, making the *reality gap* a central concern [24]. Domain randomization over cloth and environment parameters is therefore essential for developing robust policies [23, 27].

The thesis pursues the following objectives:

1. **Conveyor-based sorting task—single agent (PG-1):** Extend the validated pick-and-place pipeline from the project thesis to a full sorting task: selective retrieval of target items (T-shirts) from a mixed waste stream on a moving conveyor while ignoring non-target objects. Train Proximal Policy Optimization (PPO) policies with selectivity encoded in the reward function and evaluate sorting accuracy, false-pick rate, and missed-pick rate.
2. **Cloth simulation integration (PG-2):** Integrate deformable cloth simulation into the Isaac Sim environment using PhysX particle cloth (Position-Based Dynamics (PBD)) or

- Newton thin deformables (VBD) [3, 12]. Model T-shirt objects with realistic material properties and validate cloth behavior through qualitative and quantitative benchmarks. Apply domain randomization over cloth parameters to support robust policy learning.
3. **Second robot arm integration (PG-3):** Add a second collaborative robot arm to the simulation scene for cloth stretching and presentation. Define the arm’s kinematic configuration, placement, and workspace to complement the tendon-driven picker. Implement the stretching primitive: the second arm grasps a garment edge and extends it to a flat, camera-visible configuration.
 4. **Camera-based damage classification (PG-4):** Integrate a simulated camera system for garment inspection. Develop or integrate a classification model that assesses textile damage degree (e.g., intact, minor damage, major damage, unusable) from the stretched cloth image. Train or fine-tune the classifier on synthetic data with domain randomization.
 5. **Multi-agent coordination (PG-5):** Formulate the full pipeline as a multi-agent MDP: the tendon-driven robot picks and hands off, the sorting arm stretches, inspects, and sorts. Design the multi-agent observation and action spaces, inter-agent communication or shared reward structure, and coordination protocol. Train coordinated policies and evaluate end-to-end sorting performance.
 6. **Sim-to-real transfer preparation (PG-6):** Assess transfer robustness through systematic domain randomization (conveyor speed, cloth properties, sensor noise, lighting) [23, 27]. Define the Robot Operating System (ROS) 2 integration architecture for physical deployment: joint state publishers, policy inference nodes, camera pipelines, and safety constraints. Identify and document remaining gaps for real-world operation.
 7. **Evaluation and benchmarking (PG-7):** Define and apply comprehensive evaluation metrics: sorting accuracy by damage class, cycle time, false/missed classification rates, cloth coverage during inspection, and multi-agent coordination efficiency. Compare against single-agent baselines and ablation studies (e.g., without stretching, without classification).

1.3 Thesis Outline

This thesis is structured as follows. Section 2 provides the background required for this work, beginning with a summary of the foundation established in the project thesis (tendon-driven robot model, simulation environment, and RL pipeline) and then covering the additional topics central to the master thesis: cloth and deformable object simulation, multi-agent RL, vision-based classification, prior work on robotic cloth manipulation and sorting, and sim-to-real transfer strategies.

Section 3 identifies the specific research gap: the absence of an integrated, simulation-based multi-agent system that combines tendon-driven manipulation, deformable cloth handling, camera-based condition classification, and coordinated sorting within a single RL-trainable environment.

Section 4 describes the methodological approach, including the cloth simulation setup, the second robot arm integration, the damage classification pipeline, the multi-agent MDP formulation, the RL training procedures, and the domain randomization strategy for sim-to-real robustness.

Section 5 presents the experimental results across all objectives: single-agent sorting performance, cloth simulation fidelity, multi-agent coordination metrics, classification accuracy, and the end-to-end sorting evaluation.

Section 6 discusses the implications and limitations of the approach, including cloth simulation fidelity and its effect on policy transfer, the effectiveness of multi-agent coordination versus sequential single-agent execution, and the remaining challenges for real-world deployment.

Section 7 concludes the thesis, summarizes the contributions, and identifies directions for future work, including physical deployment via the ROS 2 integration pathway.

2 Background

2.1 Summary of Foundation (Project Thesis)

2.2 Cloth and Deformable Object Simulation

Deformable object simulation relaxes the rigid-body assumption by allowing the object’s shape to change under load. This introduces two coupled complexities: the state dimension increases dramatically (many vertices/particles instead of a single rigid transform), and contacts become configuration-dependent because deformation continuously alters the set of potential contact points and normals [8, 13]. For textiles, this coupling is particularly pronounced: applied forces not only translate the object but also create folds, wrinkles, and contact-rich self-interactions that alter future dynamics. These properties are repeatedly identified as key contributors to the reality gap in robotic cloth manipulation [8].

Common modelling approaches. Three modelling families are most relevant in robotics simulation. First, mass–spring(-damper) models discretize cloth as a mesh whose vertices are connected by springs encoding stretch and (optionally) bending penalties; such models can be made stable with implicit integration techniques but require careful parameterization and can be numerically stiff [1]. Second, particle-based constraint methods represent cloth by particles and enforce desired behavior via geometric (holonomic) constraints, which can be solved iteratively with high robustness under collisions; PBD and its variants are canonical examples [9, 12]. Third, continuum formulations discretize elastic energy and constitutive laws (e.g., in Finite Element Method (FEM)) or use hybrid particle–grid methods (e.g., Material Point Method (MPM)), providing a pathway to parameters with clearer physical interpretation at increased computational cost [5, 25].

Challenges specific to cloth dynamics. Cloth exhibits high effective DoF, frequent self-collision, and strong sensitivity to material parameters such as bending stiffness, friction, and damping. Robust self-contact and friction handling is widely recognized as a hard requirement for stable cloth simulation, particularly under large deformations and repeated collisions [2, 8]. From a learning perspective, this sensitivity implies that small modelling errors (e.g., in friction or damping) can change grasp success rates, sliding behavior, and fold formation, which in turn can qualitatively alter the learned policy’s strategy and its transferability to hardware [8, 24].

2.2.1 PBD—PhysX Particle Cloth

PhysX provides particle-based simulation through `PxBDParticleSystem`, whose solver is explicitly described as using PBD for particle–particle dynamics and as supporting phenomena including cloth, inflatables, and other deformable behaviors [15, 16]. In Omniverse, particle simulation is GPU-accelerated and is documented as supporting fluids, granular media, and cloth, including interaction with rigid bodies and articulations [20]. At the feature level, particle cloth is treated as a particle-based PBD mass–spring-style system and is marked as deprecated

in favor of newer deformable-body features, which highlights that “cloth” in Isaac Sim can refer to multiple, evolving solver families rather than a single canonical model [20, 21].

Time-step structure and iterative projection. In standard PBD, the deformable is represented by particle positions $\mathbf{x}_i \in \mathbb{R}^3$ with masses m_i ; desired behavior is enforced by a set of constraints $C_j(\mathbf{x})$ (equalities or inequalities), typically solved by iterative projection in a Gauss–Seidel fashion [12]. A schematic time step consists of: (i) updating velocities under external forces, (ii) predicting positions by an explicit step, (iii) iteratively projecting predicted positions to reduce constraint violations, and (iv) updating velocities from the corrected positions (and applying damping if configured) [12]. For a constraint $C_j(\mathbf{x}) = 0$, linearization yields a correction direction based on the constraint gradient. Using the inverse mass matrix $\mathbf{M}^{-1}$ (diagonal in the particle basis), one common update can be written as [9, 12]:

$$\Delta \mathbf{x} = k_j \cdot s_j \cdot \mathbf{M}^{-1} \nabla C_j(\mathbf{x}), \quad s_j = \frac{-C_j(\mathbf{x})}{\nabla C_j(\mathbf{x})^\top \mathbf{M}^{-1} \nabla C_j(\mathbf{x})}, \quad (1)$$

where $k_j \in [0, 1]$ is a user-defined stiffness-like scaling applied per constraint.

Effective stiffness and dependence on h and iteration count. A central practical implication of the iterative projection view is that the *effective* stiffness of constraints depends on numerical settings. Increasing the number of projection iterations typically yields stiffer constraint satisfaction at fixed h , while increasing h tends to reduce stability and can require either more iterations or smaller substeps to maintain comparable behavior [9, 10, 12]. This is beneficial for robustness and controllability, but it complicates mapping simulator parameters to physically measured material properties (e.g., bending stiffness in SI units), which is a recurring issue when calibrating cloth simulation for sim-to-real transfer [9, 24].

XPBD compliance formulation. Extended Position-Based Dynamics (XPBD) extends PBD by introducing a compliance parameter and an explicit Lagrange multiplier state to reduce time-step dependence and recover meaningful constraint force estimates [9]. For compliance $\alpha \geq 0$ and time step h , the incremental multiplier update for a constraint can be expressed as [9]:

$$\Delta \lambda = \frac{-C(\mathbf{x}) - \frac{\alpha}{h^2} \lambda}{\sum_i w_i \|\nabla_{\mathbf{x}_i} C(\mathbf{x})\|^2 + \frac{\alpha}{h^2}}, \quad \Delta \mathbf{x}_i = w_i \nabla_{\mathbf{x}_i} C(\mathbf{x}) \Delta \lambda, \quad (2)$$

with inverse masses $w_i = \frac{1}{m_i}$. This formulation decouples the user parameter α from the iteration count in a way that improves interpretability of stiffness-like behavior across different h and solver settings [9].

Why PBD is used in PhysX for particle cloth. PBD is well-aligned with GPU execution because constraints can be partitioned into independent sets and processed in parallel; graph coloring is a common strategy to schedule constraint batches without write conflicts on shared particles [4]. Omniverse documentation explicitly emphasizes GPU-accelerated PBD particles and their interaction with rigid bodies and articulations, which is essential for cloth manipulation scenes in robotics [20].

2.2.2 VBD—Newton Thin Deformables

While PhysX particle cloth is based on explicit prediction and iterative position projection, implicit methods aim to improve stability under stiffness and large time steps by solving a time-discrete

optimization problem. VBD is introduced as a block coordinate descent solver for the variational form of implicit Euler time integration, targeting convergence to the implicit Euler solution while maintaining unconditional stability under limited iteration budgets [3].

Variational implicit Euler and global energy minimization. Let x_t denote positions, v_t velocities, h the time step, and M the mass matrix. In the variational formulation, the next positions are obtained by minimizing a global energy that combines inertia and potential energy [3]:

$$x_{t+1} = \arg \min_x G(x), \quad G(x) = \frac{1}{2h^2} \|x - y\|_M^2 + E(x), \quad (3)$$

where $y = x_t + hv_t + h^2 a_{\text{ext}}$ and $E(x)$ is the total potential energy. Velocities follow from the implicit Euler relationship $v_{t+1} = \frac{x_{t+1} - x_t}{h}$ [3].

Vertex-level block coordinate descent and local Newton step. VBD solves the global problem by iterating over vertices and updating one vertex at a time while temporarily holding the others fixed, yielding a vertex-level Gauss–Seidel pattern with favorable parallelization via vertex coloring [3]. For vertex i , a local energy is minimized:

$$x_i \leftarrow \arg \min_{x_i} G_i(x), \quad G_i(x) = \frac{m_i}{2h^2} \|x_i - y_i\|^2 + \sum_{j \in F_i} E_j(x), \quad (4)$$

where F_i is the set of force/energy elements incident to vertex i . Because G_i has only three degrees of freedom, a local Newton step is formed by solving a 3×3 linear system per vertex [3]:

$$H_i \Delta x_i = f_i, \quad (5)$$

with H_i the local Hessian and f_i the local gradient/force term.

Advantages for stiff, contact-rich thin deformables. The primary advantages of VBD are: (i) unconditional stability inherited from the energy descent property, (ii) convergence toward the implicit Euler solution as iterations increase, and (iii) GPU-oriented parallelization via vertex coloring with small, localized linear solves [3]. These properties are attractive for cloth-like thin deformables, which often exhibit near-inextensible stretch (high stiffness) combined with softer bending and frequent contact events [3].

Newton in the robotics simulation ecosystem. Newton is presented as an open-source, GPU-accelerated physics engine designed for robotics simulation and robot learning, built on NVIDIA Warp and intended to integrate with robot learning frameworks such as IsaacLab [14, 17]. In NVIDIA’s robotics communication, Newton is explicitly associated with thin-deformable simulation via a VBD solver and with multiphysics capabilities that include implicit MPM and other solver components [17, 18].

2.2.3 Comparison: PBD vs. VBD for Cloth in RL Workflows

Mathematical targets and implications. The core distinction is the optimization target. PBD advances dynamics by predicting positions and then *projecting* them to reduce constraint violations, where the resulting behavior depends on solver scheduling and iteration count, and where constraint “stiffness” is typically not time-step invariant without extensions [9, 12]. VBD, in contrast, directly minimizes the variational implicit Euler energy and therefore targets

a principled implicit integration objective; stability is maintained by construction via energy descent [3].

Practical tradeoffs for learning-centric simulation. For RL, PBD-based particle cloth is attractive because it is simple, robust under frequent contacts, and aligned with GPU-parallel execution [4, 11, 20]. However, because effective stiffness depends on h and iteration count, parameter tuning can be difficult to interpret physically and can amplify sim-to-real sensitivity [9, 24]. VBD is attractive when stability under stiffness, large time steps, and mixed constraint types is prioritized [3, 18].

Relation to the Omniverse ecosystem and thesis choice. In Omniverse/Isaac Sim, particle cloth is documented as deprecated in favor of newer deformable-body features [20, 21]. For the experiments in this thesis, the PhysX/PBD particle cloth pathway is motivated by GPU throughput and robustness for large-scale training in Isaac Sim 5.1, whereas Newton/VBD is conceptually aligned with scenarios where greater stability under stiff cloth behavior and larger time steps is required [3, 18, 20].

Rationale for an integrated cloth–rigid simulation stack. Integrated cloth and deformable dynamics are supported within the Isaac Sim / PhysX stack through GPU-accelerated particle-based simulation that can represent cloth-like behavior and interact with articulations and other physics objects within a shared simulation scene [20]. In addition, IsaacLab explicitly highlights support for deformable objects (including cloth) within a framework built on Isaac Sim and designed for multi-modal learning [22]. This integration is beneficial because it avoids ad-hoc coupling between an external deformable simulator and a separate rigid-body engine, which would otherwise require custom synchronization, contact handling, and state transfer between solvers.

2.3 Multi-Agent Reinforcement Learning

2.4 Vision-Based Damage Classification

2.5 Prior Work on Cloth Manipulation

2.6 Sim-to-Real Transfer

3 Problem Statement and Research Gap

3.1 Critical Analysis of State of the Art

3.2 Research Gap and Justification

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4.4 Second Robot Arm Integration

4.5 Camera-Based Damage Classification

4.6 Multi-Agent RL Formulation

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5.2 Cloth Simulation Results

5.3 Multi-Agent Sorting Pipeline Results

5.4 Damage Classification Results

5.5 Ablation Studies

5.6 Sim-to-Real Readiness Assessment

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6.2 Comparison with Related Work

6.3 Limitations

6.4 Practical Implications

7 Conclusion

7.1 Conclusion

7.2 Outlook

Bibliography

- [1] BARAFF, David and WITKIN, Andrew. Large Steps in Cloth Simulation. In : *Proceedings of the 25th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '98)*. 1998. p. 43–54. DOI [10.1145/280814.280821](https://doi.org/10.1145/280814.280821).
- [2] BRIDSON, Robert, FEDKIW, Ronald and ANDERSON, John. Robust Treatment of Collisions, Contact and Friction for Cloth Animation. In : *Proceedings of the 29th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '02)*. 2002. p. 594–603. DOI [10.1145/566570.566623](https://doi.org/10.1145/566570.566623).
- [3] CHEN, Anka He, LIU, Ziheng, YANG, Yin and YUKSEL, Cem. Vertex Block Descent. *ACM Transactions on Graphics (Proceedings of SIGGRAPH)*. 2024. Vol. 43, no. 4. DOI [10.1145/3658179](https://doi.org/10.1145/3658179).
- [4] FRATARCANGELI, Marco and PELLACINI, Fabio. A GPU-Based Implementation of Position Based Dynamics for Interactive Deformable Bodies. *Journal of Graphics Tools*. 2013. Vol. 17, no. 3, p. 59–66. DOI [10.1080/2165347X.2015.1030525](https://doi.org/10.1080/2165347X.2015.1030525).
- [5] JIANG, Chenfanfu, SCHROEDER, Craig, TERAN, Joseph, STOMAKHIN, Alexey and SELLE, Andrew. The Material Point Method for Simulating Continuum Materials. *ACM Transactions on Graphics*. Online. 2016. Vol. 35, no. 4. DOI [10.1145/2897826.2927348](https://doi.org/10.1145/2897826.2927348).
- [6] KLEIN, Max. *Arbeitsraumanalyse, Simulation und Bewegungsplanung eines seilgetriebenen robotischen Manipulators*. Masterarbeit. 2023.
- [7] KOBER, Jens, BAGNELL, J. Andrew and PETERS, Jan. Reinforcement Learning in Robotics: A Survey. *The International Journal of Robotics Research*. 2013. Vol. 32, no. 11, p. 1238–1274. DOI [10.1177/0278364913495721](https://doi.org/10.1177/0278364913495721).
- [8] LONGHINI, Alberta, WANG, Yufei, GARCIA-CAMACHO, Irene, BLANCO-MULERO, David, MOLETTA, Marco, WELLE, Michael, ALENYÀ, Guillem, YIN, Hang, ERICKSON, Zackory, HELD, David, BORRÀS, Júlia and KRAGIC, Danica. A Review of Robotic Cloth Manipulation. *Annual Review of Control, Robotics, and Autonomous Systems*. 2025. DOI [10.1146/annurev-control-022723-033252](https://doi.org/10.1146/annurev-control-022723-033252).
- [9] MACKLIN, Miles, MÜLLER, Matthias and CHENTANEZ, Nuttapong. XPBD: Position-Based Simulation of Compliant Constrained Dynamics. *Proceedings of the 9th International Conference on Motion in Games (MIG)*. 2016. DOI [10.1145/2994258.2994272](https://doi.org/10.1145/2994258.2994272).
- [10] MACKLIN, Miles, STOREY, Kier, LU, Michelle, TERDIMAN, Pierre, CHENTANEZ, Nuttapong, JESCHKE, Stefan and MÜLLER, Matthias. Small Steps in Physics Simulation. Online. 2019. [Accessed 3 March 2026]. Available from: <https://mmacklin.com/smallsteps.pdf>
- [11] MAKОВИYCHUK, Viktor, WAWRZYNIAK, Lukasz, GUO, Yunrong, LU, Michelle, STOREY, Kyle, MACKLIN, Miles, HOELLER, David, RUDIN, Nikita, ALLSHIRE, Arthur, HANDA, Ankur, STATE, Gavriel, GIRDHAR, Ankur, KUMAR, Tushar, FINK, Jonathan, MEREL, Josh and TASSA, Yuval. Isaac Gym: High Performance GPU Based Physics Simulation

- for Robot Learning. In : *NeurIPS 2021 Datasets and Benchmarks Track (Round 2)*. Online. 2021. Available from: https://openreview.net/forum?id=fgFBtYgJQX_
- [12] MÜLLER, Matthias, HEIDELBERGER, Bruno, HENNIX, Marcus and RATCLIFF, John. Position Based Dynamics. In : *Journal of Visual Communication and Image Representation*. 2007. p. 109–118. DOI [10.1016/j.jvcir.2007.01.005](https://doi.org/10.1016/j.jvcir.2007.01.005).
- [13] NEALEN, Andrew, MÜLLER, Matthias, KEISER, Richard, BOXERMAN, Eddy and CARLSON, Mark. Physically Based Deformable Models in Computer Graphics. *Computer Graphics Forum*. 2006. Vol. 25, no. 4, p. 809–836. DOI [10.1111/j.1467-8659.2006.01000.x](https://doi.org/10.1111/j.1467-8659.2006.01000.x).
- [14] NEWTON PHYSICS CONTRIBUTORS. Newton Physics Documentation. Online. 2025. [Accessed 3 March 2026]. Available from: <https://newton-physics.github.io/newton/>
- [15] NVIDIA. Particle System — PhysX 5.1.3 Documentation. Online. 2023. [Accessed 3 March 2026]. Available from: <https://nvidia-omniverse.github.io/PhysX/physx/5.1.3/docs/ParticleSystem.html>
- [16] NVIDIA. PxPBDParticleSystem — PhysX API Reference. Online. 2023. [Accessed 3 March 2026]. Available from: https://nvidia-omniverse.github.io/PhysX/physx/5.3.1/_api_build/class_px_p_b_d_particle_system.html
- [17] NVIDIA. Announcing Newton, an Open-Source Physics Engine for Robotics Simulation. Online. 2025. [Accessed 3 March 2026]. Available from: <https://developer.nvidia.com/blog/announcing-newton-an-open-source-physics-engine-for-robotics-simulation/>
- [18] NVIDIA. Train a Quadruped Locomotion Policy and Simulate Cloth Manipulation with NVIDIA Isaac Lab and Newton. Online. 2025. [Accessed 3 March 2026]. Available from: <https://developer.nvidia.com/blog/train-a-quadruped-locomotion-policy-and-simulate-cloth-manipulation-with-nvidia-isaac-lab-and-newton/>
- [19] NVIDIA. What Is Isaac Sim?. Online. 2026. [Accessed 3 March 2026]. Available from: <https://docs.isaacsim.omniverse.nvidia.com/5.1.0/index.html> Isaac Sim Documentation (v5.1.0), last updated Jan 28, 2026
- [20] NVIDIA. Particles. Online. 2026. [Accessed 3 March 2026]. Available from: https://docs.omniverse.nvidia.com/kit/docs/omni_physics/latest/dev_guide/particles/particles.html Omni Physics documentation, last updated Jan 20, 2026
- [21] NVIDIA. Omniverse Physics and PhysX SDK Limitations — Isaac Sim 5.1.0 Documentation. Online. 2026. [Accessed 3 March 2026]. Available from: https://docs.isaacsim.omniverse.nvidia.com/5.1.0/physics/physics_resources.html
- [22] NVIDIA, MITTAL, Mayank, ROTH, Pascal, TIGUE, James, RICHARD, Antoine, ZHANG, Octi, DU, Peter, SERRANO-MUÑOZ, Antonio, YAO, Xinjie, ZURBRÜGG, René, RUDIN, Nikita, WAWRZY尼亚K, Lukasz and OTHERS. Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning. Online. 2025. arXiv:2511.04831
- [23] PENG, Xue Bin, ANDRYCHOWICZ, Marcin, ZAREMBA, Wojciech and ABBEEL, Pieter. Sim-to-Real Transfer of Robotic Control with Dynamics Randomization. In : *2018 IEEE International Conference on Robotics and Automation (ICRA)*. 2018. p. 3803–3810. DOI [10.1109/ICRA.2018.8460528](https://doi.org/10.1109/ICRA.2018.8460528).

- [24] SALVATO, Erica, FENU, Gianfranco, MEDVET, Eric and PELLEGRINO, Felice Andrea. Crossing the Reality Gap: A Survey on Sim-to-Real Transferability of Robot Controllers in Reinforcement Learning. *IEEE Access*. 2021. Vol. 9, p. 153171–153187. DOI [10.1109/ACCESS.2021.3126658](https://doi.org/10.1109/ACCESS.2021.3126658).
- [25] SIFAKIS, Eftychios and BARBIČ, Jernej. FEM Simulation of 3D Deformable Solids: A Practitioner's Guide to Theory, Discretization and Model Reduction. In : *ACM SIGGRAPH 2012 Courses*. 2012. DOI [10.1145/2343483.2343501](https://doi.org/10.1145/2343483.2343501).
- [26] SUTTON, Richard S. and BARTO, Andrew G. *Reinforcement Learning: An Introduction*. Online. Second. The MIT Press, 2018. [Accessed 3 March 2026]. ISBN 9780262039246. Available from: <https://mitpress.mit.edu/9780262039246/reinforcement-learning/>
- [27] TOBIN, Josh, FONG, Rachel, RAY, Alex, SCHNEIDER, Jonas, ZAREMBA, Wojciech and ABBEEL, Pieter. Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World. In : *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Online. 2017. DOI [10.1109/IROS.2017.8202133](https://doi.org/10.1109/IROS.2017.8202133).

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